Tuberculosis: An Approach to a Case of Haemoptysis

Dr. Essam Kotb, Internal Medicine | Source: `12- Tuberculosis updated.pdf` (38 slides)

Legend: ⭐ = high-yield · *[added]* = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes. The two case answers are **reasoned by me; the key is not on the slides**.

🔗 **Bridge from earlier lectures:**

- **Pneumonia (Kotb):** TB is the "pneumonia that doesn't respond". Suspect it with cough > 2–3 weeks, weight loss, night sweats, or an upper-lobe cavity.
- **Lung abscess / bronchiectasis:** both are **cavitating / haemoptysis** differentials. Post-TB bronchiectasis is a classic upper-lobe cause.
- **Lung cancer:** the other big cause of **haemoptysis + weight loss** in a smoker.
- **Pleural diseases: TB effusion** = **lymphocytic exudate**, high **ADA**; pleural biopsy.
- **Pericardial disease:** TB → effusion and **constrictive pericarditis**.
- **Adrenal II: TB adrenalitis = 10–20% of primary Addison's**.
- **Infective endocarditis / psychiatry:** isoniazid hepatitis, rifampicin enzyme induction and interactions *[added link]*.

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1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	Epidemiology, reservoirs, transmission of the **M. tuberculosis complex**	M. bovis → cow's milk
Knowledge	⭐ **Primary TB, LTBI, reactivation**, tissue destruction	Ghon complex; upper-lobe reactivation
Knowledge	Pulmonary and **extrapulmonary** presentations	Pott, miliary, TB meningitis, GU TB
Knowledge	⭐ Diagnostics and regimens: latent, active, drug-susceptible, drug-resistant	RIPE 2+4; 4-month HPZM; 3HP
Skill	Interpret history, exam, labs, imaging	TST/IGRA choice
Skill	Initiate the intensive and continuation phases; manage **interruptions**; adjuncts	≥ 14-day gap → restart
Values	Professionalism; ⭐ **public health**: mandatory reporting, airborne precautions, contact tracing	Notification

Main subtopics:

- aetiology
- classification
- pathogenesis
- clinical manifestations
- diagnostics
- treatment

- subtypes/variants
- complications

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2. Main Content (Lecture Order)

2.1 Definition (Slide 5)

- **TB** = disease caused by ***Mycobacterium tuberculosis***, typically affecting the **lungs**.
- **Primary infection: airborne aerosol (droplet)** transmission; often **initially asymptomatic**.
- **Latent infection:** bacilli are **dormant** because innate and cellular immunity are intact.
- **Reactivation:** if immunity is compromised, latent infection reactivates → **active TB**.

Definition:
A disease caused by *Mycobacterium tuberculosis*, which typically affects the lungs.

Primary infection, transmitted via airborne aerosol (droplet), is often initially asymptomatic.

Latent infection, *M. tuberculosis* infection is typically dormant, because of intact innate and cellular immune responses. If the immune system is compromised, reactivation of the infection may occur and lead to active TB.



Slide 5: Lung infected with TB

2.2 Aetiology: the M. tuberculosis Complex (Slide 6)

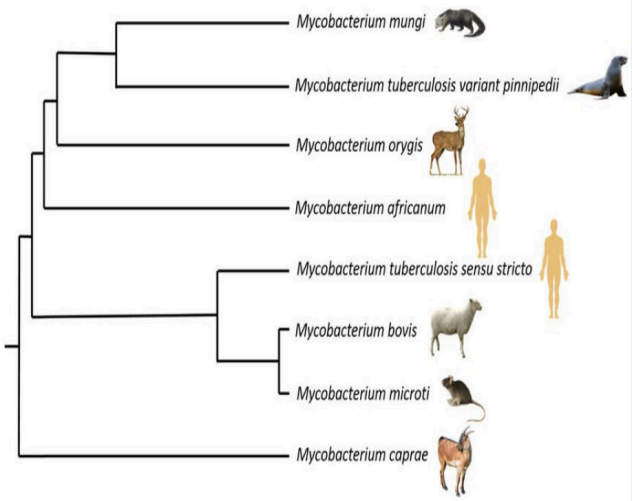
Species	Transmission	Reservoir / note
⭐ ***M. tuberculosis***	**Aerosol droplet nuclei**	**Humans**
⭐ ***M. bovis***	⭐ **Ingestion of contaminated cow's milk** (→ GI TB)	**Cattle** (*[added]* also the source of the **BCG** vaccine strain)
M. africanum	—	Common cause of TB in **West, Central, East Africa**
M. microti	—	(Rodents; rarely human)

Etiology

Mycobacteria Species

Mycobacterium species that cause tuberculosis are collectively known as the *Mycobacterium tuberculosis* complex, which includes:

- ***Mycobacterium tuberculosis***
  - Mode of transmission: spread via aerosol droplet nuclei
  - Reservoir: predominantly humans
- ***Mycobacterium bovis***
  - Mode of transmission: ingestion of contaminated cow's milk
  - Reservoir: predominantly cattle
- ***Mycobacterium africanum***: common cause of tuberculosis in West, Central, and East Africa
- ***Mycobacterium microti***



Slide 6: The M. tuberculosis complex

2.3 Risk Factors for Exposure (Slide 7)

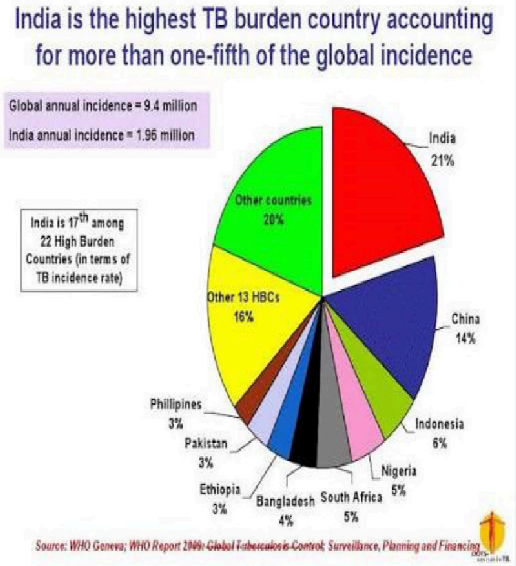
- **Working in health care**
- **Migration from** or **frequent travel to** high-incidence countries
- ⭐ **Close contact** with active TB
- **Crowded living** (prisons); **homelessness**
- **Smoking**
- ⭐ **Associated disease: HIV, renal failure, DM**

Epidemiology on the slide: India is the highest-burden country, **> one-fifth (21%) of global incidence**; global incidence **9.4 million**, India 1.96 million.

- ⚠️ **These are 2009 WHO figures.**
- *[added]* WHO Global TB Report 2024 (2023 data): **~10.8 million** new cases; **India ~26%**; TB is again the **leading single infectious killer**.

Risk factors for tuberculosis exposure

- Working in the health care industry
- Migration from countries with a high TB incidence
- Frequent travel to countries with a high TB burden
- Close contact with a patient with active TB infection
- Crowded living conditions (e.g., prisons)
- Homelessness
- Smoking
- Associated disease (HIV, Renal failure, DM)



Slide 7: Risk factors; global burden

2.4 Pathophysiology (Slides 8–11) ⭐⭐

Primary TB: innate response (slide 8)

1. Inhaled droplet nuclei reach the **terminal alveoli**.
2. Taken up by **alveolar macrophages** (phagocytosis).
3. ⭐ ***M. tuberculosis* survives within macrophages** (*[added]* it blocks phagolysosome fusion).
4. **Macrophage lysis** releases bacilli, which infect new macrophages.
5. Some enter the **bloodstream → bacteraemia** → seeding of multiple organs.

Granulomatous inflammation (slide 9)

- Destruction of infected macrophages → ⭐ **central CASEOUS necrosis**.
- The **granuloma limits spread**.
- ⭐ **Ghon focus:** a granuloma, typically in the **middle/lower lobes**.
- ⭐ **Ghon complex: Ghon focus + regional (hilar) lymph node + the linking lymphatics**.
- *[added]* **Ranke complex** = a calcified Ghon complex.

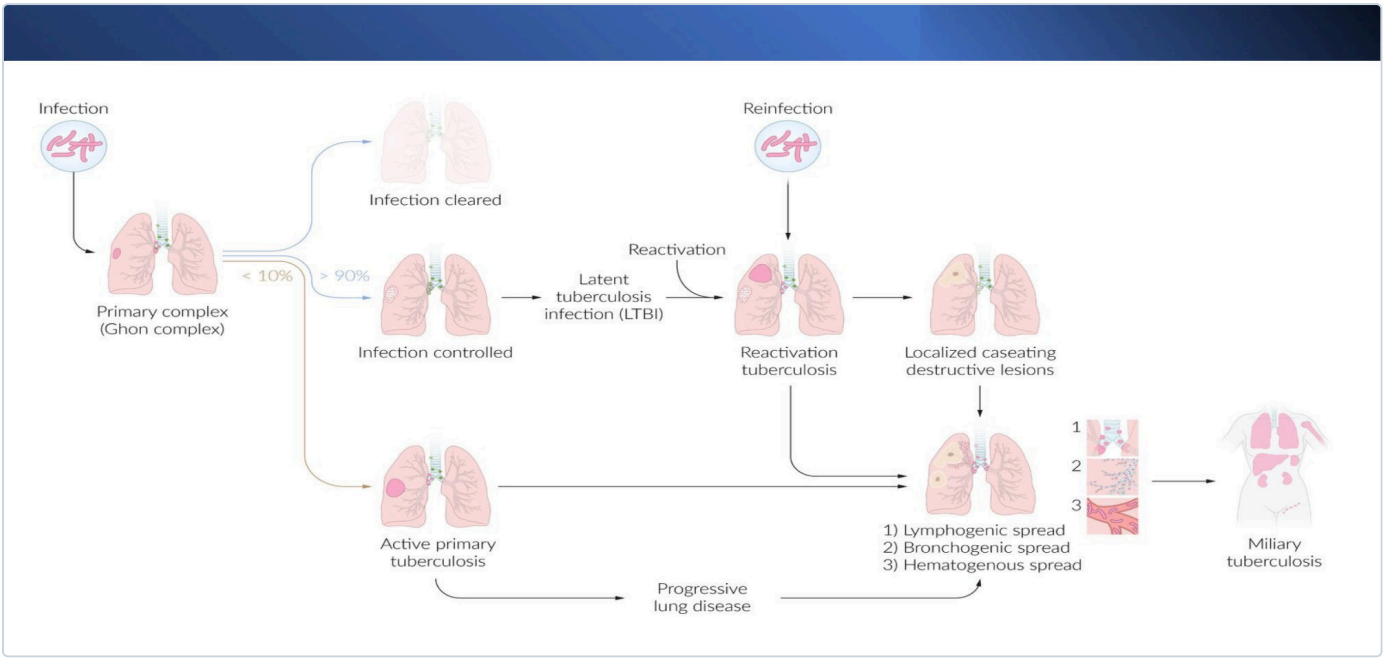
Immune response	Outcome
Sufficient	Most bacilli killed; some persist → ⭐ **latent TB**; Ghon complex **fibroses and calcifies**
Deficient (HIV, malnutrition)	Granulomas fail → ⭐ **progressive primary TB** → progressive lung disease, bacteraemia, **miliary TB**

Secondary (reactivation) TB (slide 10)

- **Latent TB:** dynamic equilibrium between host immunity and bacilli.
- **Reactivation** when immunity weakens.
- ⭐ **Caseating granulomas with central necrosis and Langhans giant cells**.
- ⭐ **Upper lobes**, because of **higher oxygen tension** (*[added]* apical segments of the upper lobes, superior segment of the lower lobe).
- Prior sensitisation → **stronger inflammatory response** → ⭐ **extensive tissue destruction, cavitation, scarring**.

Natural history (slide 11):

1. Infection → **primary (Ghon) complex**.
2. Then one of:
   - **> 90%: infection controlled** → LTBI (or infection cleared). LTBI → **reactivation** (or **reinfection**) → localised caseating destructive lesions.
   - **< 10%: active primary TB** → progressive lung disease.
3. Either path can spread by **lymphogenic, bronchogenic or haematogenous** routes → **miliary TB**.



Slide 11: Natural history of TB

	Primary TB	Secondary (reactivation) TB
Who	First exposure, *[added]* children	Previously infected; immunity falls
Site	**Mid/lower zones** + hilar nodes (Ghon complex)	⭐ **Upper lobes** (high O_2)
Lesion	Ghon focus → fibrosis/calcification	**Cavitation**, caseation, scarring
Course	Usually asymptomatic; progressive if immunodeficient	Symptomatic; infectious (cavity)

🔁 **Recap: Pathogenesis.**

- Droplet → alveolar macrophage (bacilli survive inside) → caseating granuloma → Ghon complex.
- > 90% controlled (latent); < 10% progress.
- Reactivation targets the **upper lobes** (O_2), with cavitation, Langhans giant cells, caseation.
- HIV/malnutrition → progressive primary or **miliary** TB.

▶️ **Up next: Clinical features.**

2.5 Pulmonary TB: Symptoms and Signs (Slides 12–13) ⭐

Systemic	Pulmonary
⭐ **Low-grade fever with night sweats**	**Non-productive cough → productive (purulent) cough**
⭐ **Weight loss (often severe)**, anorexia	⭐ **Haemoptysis**
Malaise, weakness	**Shortness of breath**
—	**Pleuritic chest pain**

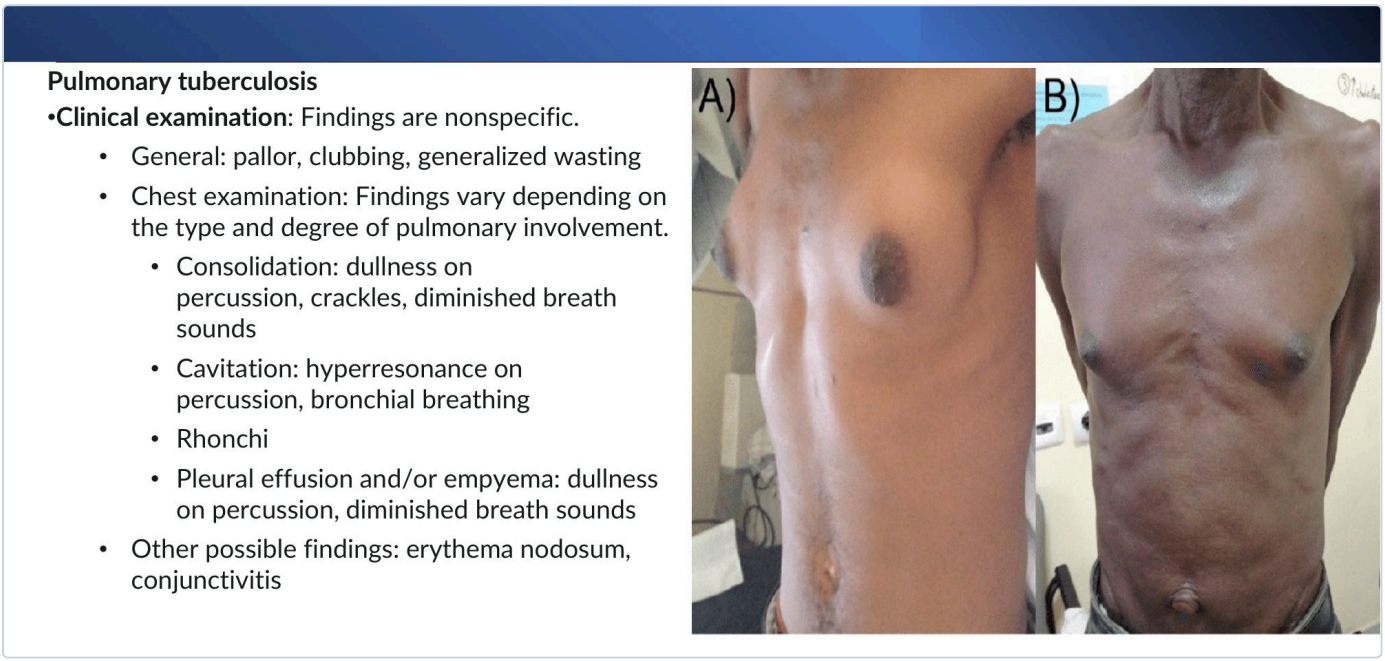
Examination (nonspecific):

- **General: pallor, clubbing, generalised wasting**.

Chest finding	Signs
Consolidation	Dullness, crackles, ↓ breath sounds

Chest finding	Signs
⭐ **Cavitation**	**Hyperresonance, bronchial breathing** (*[added]* "amphoric")
—	**Rhonchi**
Pleural effusion/empyema	Dullness, ↓ breath sounds

- **Other: ⭐ erythema nodosum**, **conjunctivitis** (*[added]* phlyctenular).
- 🔎 **Slide 13 photos (unlabelled, my reading):** A = chest-wall/axillary swelling (likely lymphadenitis or a cold abscess); B = generalised wasting.



Slide 13: Examination findings and photos

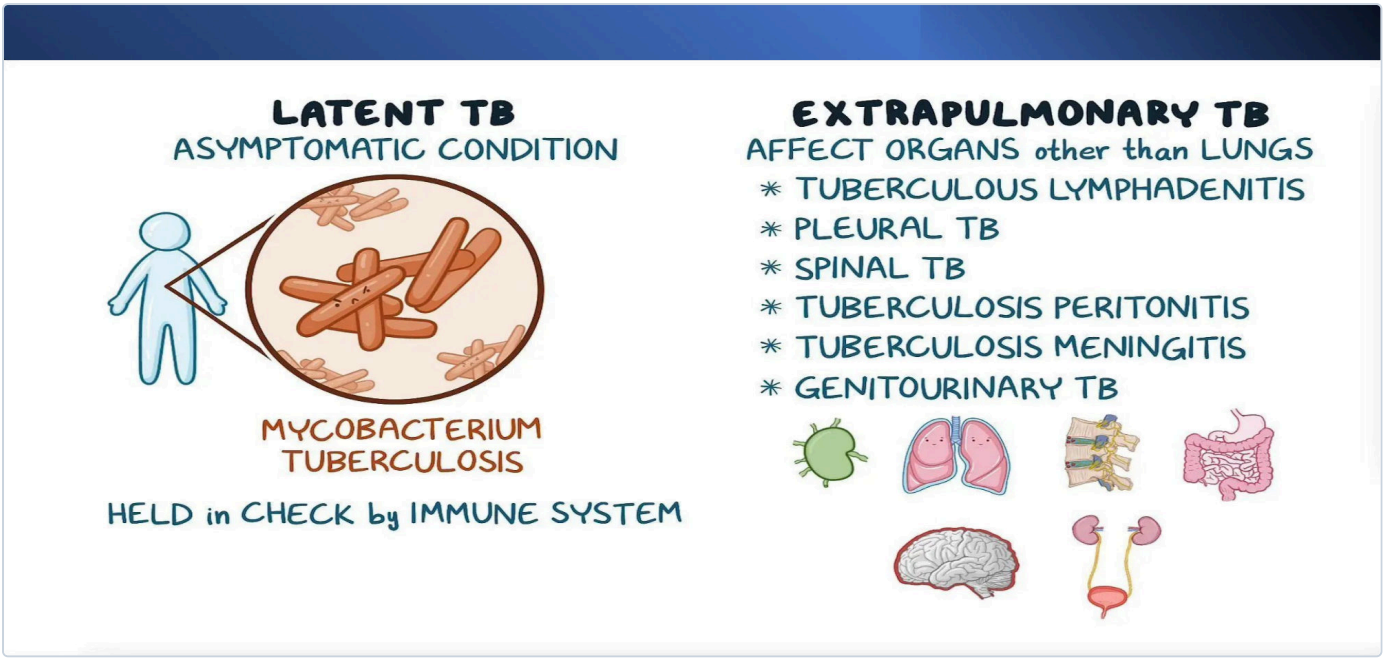
[added] **Classic triad to remember: cough > 2–3 weeks + night sweats + weight loss** → sputum for TB (and CXR).

2.6 Latent TB and Extrapulmonary TB (Slides 14–21) ⭐

Latent TB: an **asymptomatic** condition; bacilli **held in check by the immune system**.

Extrapulmonary TB: affects organs other than the lungs:

- **lymphadenitis**
- **pleural**
- **spinal**
- **peritonitis**
- **meningitis**
- **genitourinary**



Slide 14: Latent vs extrapulmonary TB

Form	Pathophysiology	Features
⭐ **TB lymphadenitis** (*[added]* "scrofula"; the **commonest** extrapulmonary form)	—	Sites: **cervical**, inguinal, axillary. **Lymphadenitis:** firm, mobile, discrete → **Periadenitis:** rubbery, **fixed** → ⭐ **Cold abscess:** soft, **fluctuant** (*[added]* → sinus)
TB pleurisy	Direct spread from lung; haematogenous; ⭐ **delayed hypersensitivity** to mycobacterial antigen in the pleural space	Constitutional symptoms, non-productive cough, **intense pleuritic pain**, dyspnoea
⭐ **Miliary TB**	**Lymph or blood spread** from a pulmonary or extrapulmonary source → multi-organ granulomas	Sites: lungs, spleen, liver, nodes, **adrenals**, **meninges**, vertebrae, joints, bones, **choroid**. Cough, dyspnoea, constitutional symptoms, lymphadenopathy, **hepatosplenomegaly**, ⭐ **choroidal tubercles on fundoscopy**. *[added]* CXR: diffuse **millet-seed** 1–3 mm nodules
Pericardial TB	—	**Effusion** (rarely **tamponade**) or ⭐ **constrictive pericarditis**; chest pain, cough, dyspnoea
Cutaneous TB	—	Diagnosis: **skin biopsy** (histopathology, AFB stain, culture)
GI TB	⭐ **Ingestion of infected milk (M. bovis) or swallowed sputum**; haematogenous/lymphatic	Peritoneum, oesophagus, stomach, jejunum, ⭐ **ileocaecum** (*[added]* mimics **Crohn's**), colorectal
Urogenital TB	Haematogenous spread to **kidneys** → ureters/bladder via urine	Flank pain, **haematuria**, storage symptoms (frequency, nocturia, urgency), **hypertension**, ⭐ **recurrent UTIs unresponsive to standard antibiotics** (*[added]* **sterile pyuria**)
⭐ **Pott disease** (spinal TB)	Haematogenous spread to **vertebral bodies** → **intervertebral disc** destruction and collapse	Back pain, spinal tenderness, ⭐ **kyphosis (gibbus)**, deformity, **neurological deficit (paraplegia)**; **cold abscess** (*[added]* psoas); fever worse at night. Lower thoracic/lumbar; children and young adults

⚠️ **Slide 18 error:** it is headed "**Tuberculous meningitis**" but its content is **pericardial TB**. TB meningitis has no clinical slide (only its treatment, slide 31).

- *[added]* **TB meningitis:** subacute headache, fever, **cranial nerve palsies (VI)**, ↓ consciousness, **basal meningitis** on MRI, hydrocephalus.
- *[added]* **CSF: lymphocytes, very high protein, very low glucose**, ADA ↑; "cobweb" clot.

Extrapulmonary tuberculosis

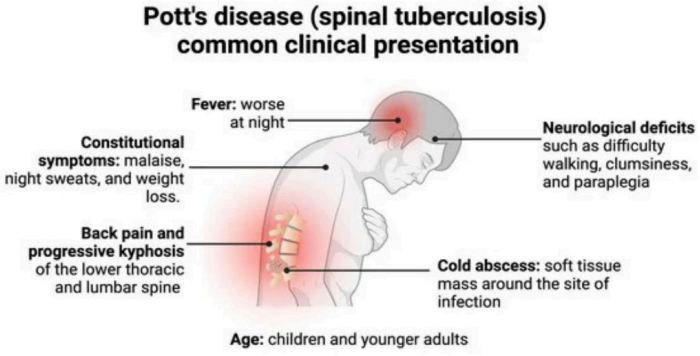
Pott disease

•Pathophysiology

- Hematogenous dissemination of infection to vertebral bodies following primary TB
- Infection from vertebral bodies spreads to the intervertebral disk, causing destruction and collapse of the disk

•Clinical features

- Constitutional symptoms
- Back pain
- Spinal tenderness
- Kyphosis
- Deformity
- Neurological deficit (e.g., paraplegia)



Slide 21: Pott disease

🔁 **Recap: Clinical.**

- Pulmonary: fever, night sweats, weight loss, cough → haemoptysis; signs of consolidation or cavity; erythema nodosum.
- Extrapulmonary:
- lymph nodes (commonest; cold abscess)
- pleura (hypersensitivity)
- miliary (choroidal tubercles)
- pericardium (constriction)
- GI (M. bovis, ileocaecal)
- GU (sterile pyuria, resistant UTIs)
- Pott (kyphosis, paraplegia)

▶️ **Up next: Diagnosis.**

2.7 Diagnosis (Slides 22–26) ⭐⭐

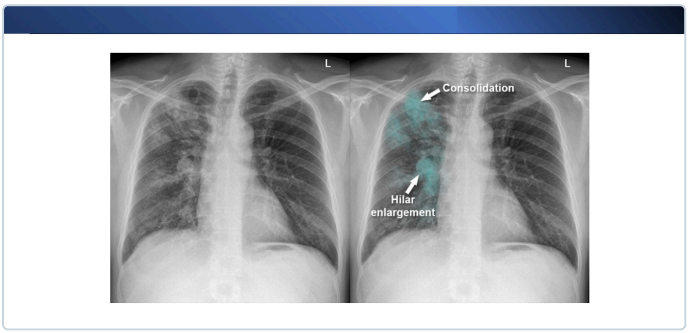
Initial work-up (slide 22):

- **History:** exposure; living in or visiting high-burden countries.
- **Microbiology:**
  - initial: ⭐ **AFB microscopy** + ⭐ **nucleic acid amplification test (NAAT)** (*[added]* e.g. **GeneXpert MTB/RIF**, which also detects **rifampicin resistance**)
  - confirmatory: ⭐ **mycobacterial culture** (the gold standard; *[added]* takes **2–8 weeks**)
  - **drug susceptibility testing** in certain circumstances
- **CT chest or CXR.**

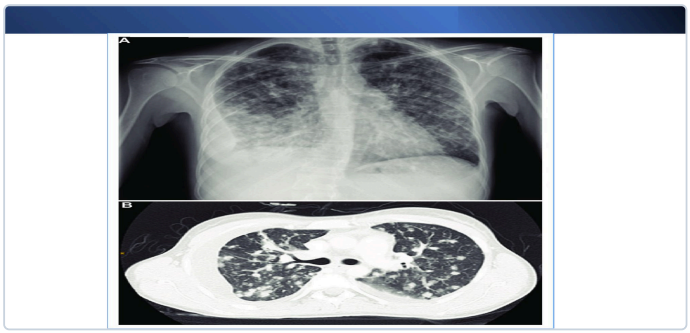
- **Suspected extrapulmonary TB:** invasive sampling (**lymph node biopsy, thoracocentesis**).

Imaging (slides 23–24):

- **Slide 23 CXR:** right upper-zone **consolidation** + **hilar enlargement** (*[added]* the pattern of **primary TB**).
- 🔎 **Slide 24 (unlabelled, my reading):**
  - **A (CXR):** bilateral, predominantly upper-zone nodular/reticulonodular opacities with volume loss
  - **B (CT):** multiple nodules with a cavity and **tree-in-bud** pattern = active **endobronchial spread** (post-primary TB)



Slide 23: CXR consolidation + hilar enlargement



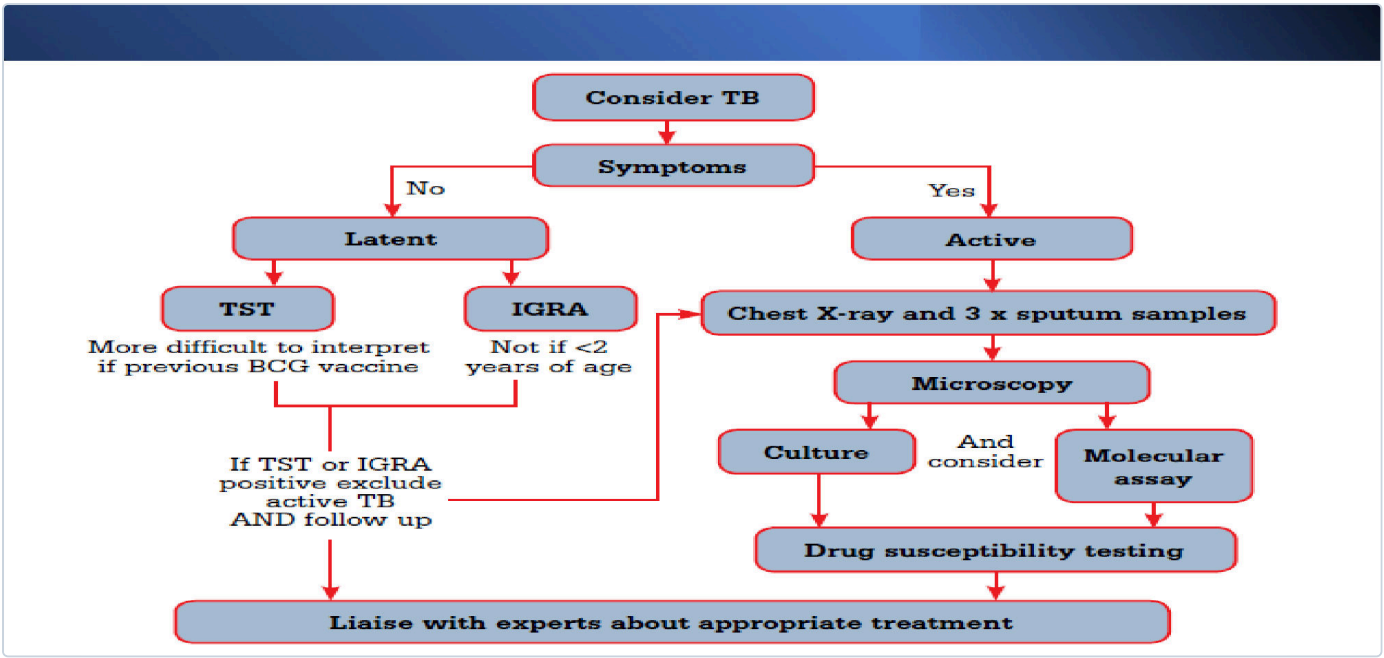
Slide 24: CXR and CT of TB

Laboratory studies (slide 25): unspecific, but useful early.

- **CBC:** mild **leukocytosis**, **anaemia**
- **Sodium:** mild **hyponatraemia** (*[added]* SIADH; or adrenal involvement)
- **Liver enzymes:** baseline before treatment
- ↑ **ESR**; **pyuria** in renal disease
- ⭐ **IGRA (QuantiFERON)**
- **Microbiology:** bacilli confirmed by **direct visualisation, culture or genetic material**.
  - ⭐ **Sputum: ≥ 3 samples, 8–24 h apart, including ≥ 1 early-morning sample**
  - **induced sputum**; **gastric lavage** (*[added]* children); **bronchoalveolar lavage**
  - **Extrapulmonary:** fluid/tissue for cell count, chemistry, AFB smear and ⭐ **adenosine deaminase (ADA)**

Diagnostic algorithm (slide 26) ⭐

```
Consider TB → Symptoms?
 ├─ NO → LATENT → TST (harder to interpret after BCG)  or  IGRA (not if < 2 years old)
 │          └─ If TST or IGRA positive → EXCLUDE ACTIVE TB (CXR) and follow up
 └─ YES → ACTIVE → CXR + 3 × sputum → microscopy → culture ± molecular assay
                  → drug susceptibility testing → liaise with experts on treatment
```



Slide 26: TB diagnostic algorithm

TST vs IGRA *[added detail; principle on slide 26]*

	TST (Mantoux)	IGRA (QuantiFERON)
Method	Intradermal PPD; read induration at **48–72 h**	Blood: IFN-γ release to TB-specific antigens (ESAT-6, CFP-10)
BCG cross-reaction	⭐ **Yes**: false positives	⭐ **No** (antigens absent from BCG)
Visits	2	1
Preferred in	**< 2 years** (IGRA less validated)	⭐ **BCG-vaccinated, ≥ 2 years**, unlikely to return for reading
Limitation (both)	**Cannot distinguish latent from active**; false negatives in immunosuppression	Same

TST positivity cut-offs *[added; CDC]*:

Induration	Positive in
≥ 5 mm	**HIV**, recent contact, **fibrotic changes on CXR**, transplant/immunosuppressed (e.g. steroids ≥ 15 mg/day for ≥ 1 month)
≥ 10 mm	Recent immigrants (< 5 years) from high-burden countries, **IV drug users**, **residents/employees of high-risk congregate settings** (prisons, nursing homes, hospitals, shelters), mycobacteriology lab staff, **DM, CKD**, silicosis, children < 4
≥ 15 mm	**No risk factors**

🔁 **Recap: Diagnosis.**

- Active TB: CXR + **3 sputum (≥ 1 early morning)** → AFB smear + **NAAT** → **culture (gold standard)** → DST.
- Extrapulmonary: tissue/fluid with **ADA**.
- Latent TB: **TST** (BCG confounds; < 2 years) or **IGRA** (BCG-vaccinated). A positive test → exclude active disease with CXR.
- Labs: anaemia, mild leukocytosis, hyponatraemia, ↑ ESR; baseline LFTs.

▶️ **Up next: Treatment.**

2.8 Treatment: General Principles (Slide 27)

Goals:

1. ↓ disease severity and **transmission**
2. **eradicate all bacilli** → sustained cure without relapse
3. ⭐ **prevent drug resistance** (*[added]* hence multi-drug therapy)

⭐ Infection control:

- **Case notification: mandatory** reporting to the local health department.
- **Airborne precautions:** a **surgical mask for the patient**; **respirators (N95)** and PPE for staff and caregivers (*[added]* negative-pressure room).
- **Contact tracing.**

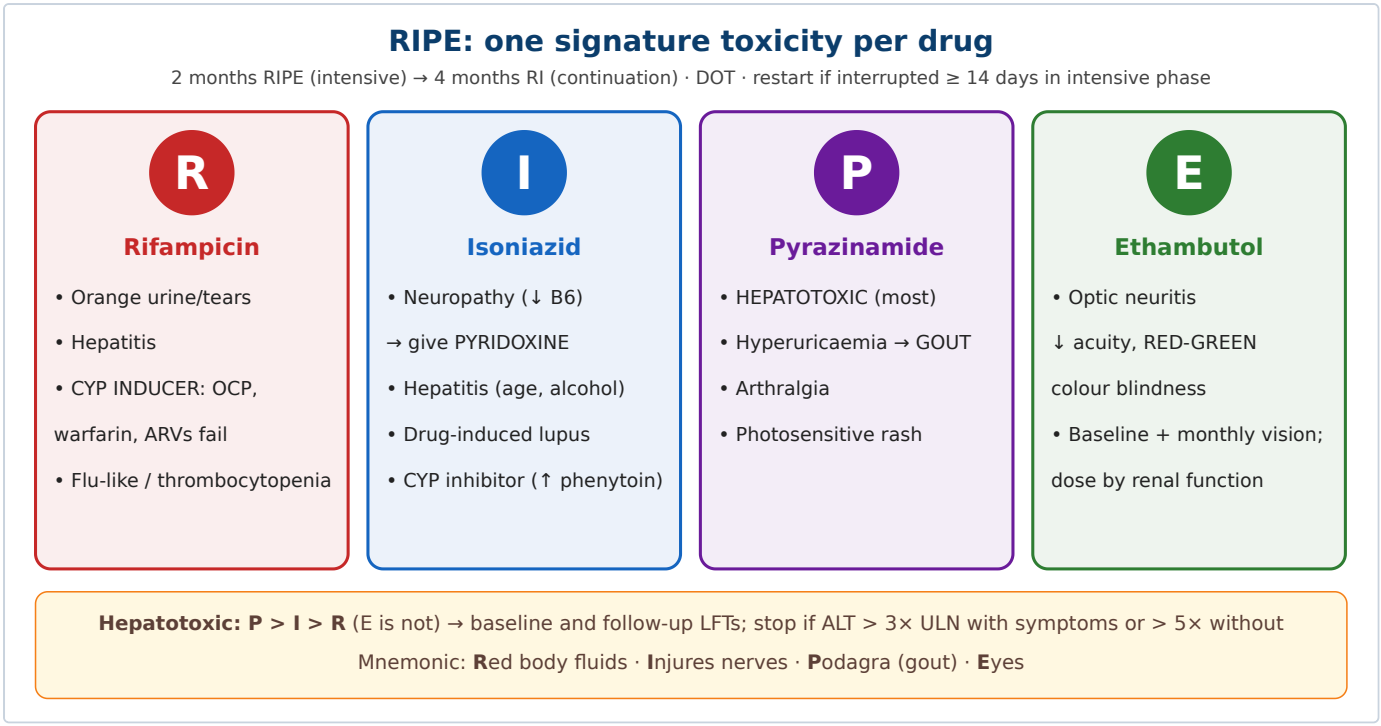
2.9 RIPE Regimen (Slide 28) ⭐⭐

Phase	Duration	Drugs
⭐ **Intensive**	**8 weeks (~2 months)**	**R**ifampicin + **I**soniazid + **P**yrazinamide + **E**thambutol
⭐ **Continuation**	**18 weeks (~4 months)**	**Rifampicin + isoniazid**
Adjunct	Throughout	⭐ **Pyridoxine (B6)** for everyone at risk of neuropathy. Isoniazid **promotes pyridoxine excretion**

- **Self-administered and daily dosing preferred.**
- ⭐ **Directly observed therapy (DOT)** is the standard of practice.
- ⚠️ The slide lists "self-administered" and "DOT standard" together, which reads as contradictory. Current guidance: **daily dosing**, with DOT or video-observed therapy as the standard where available.
- ⭐ **Interruption ≥ 14 days during the intensive phase → restart full treatment.**



Slide 28: RIPE regimen



Added diagram: RIPE signature toxicities

RIPE toxicities *[added]*:

Drug	Mechanism	Key toxicities
Rifampicin	Inhibits DNA-dependent **RNA polymerase**	**Orange-red body fluids**, hepatitis, ⭐ **CYP450 induction** (OCP, warfarin, ARVs, steroids, methadone fail), flu-like syndrome, thrombocytopenia
Isoniazid	Inhibits **mycolic acid** synthesis	⭐ **Peripheral neuropathy** (B6), **hepatitis** (↑ with age, alcohol; slow acetylators), drug-induced **lupus**, sideroblastic anaemia, CYP inhibitor
Pyrazinamide	—	⭐ **Most hepatotoxic**; **hyperuricaemia → gout**; arthralgia
Ethambutol	Inhibits arabinosyl transferase	⭐ **Optic neuritis: ↓ acuity, red-green colour blindness** (renal dosing; check vision)

2.10 Four-Month Regimen (Slide 29)

Eligibility: drug-susceptible pulmonary TB, **age ≥ 12 years, ≥ 40 kg**, no contraindications (**drug interactions, prolonged QT, pregnancy**).

Phase	Duration	Drugs
Intensive	**8 weeks**	**Rifapentine + isoniazid + pyrazinamide + moxifloxacin**
Continuation	**9 weeks**	**Rifapentine + isoniazid + moxifloxacin**
Adjunct	—	**Pyridoxine**

[added] Moxifloxacin prolongs QT, hence the QT contraindication. This is the CDC 2022 "**HPZM**" regimen (**17 weeks total**).

2.11 Drug-Resistant TB (Slide 30)

- **Oral agents:**
  - **fluoroquinolone (levofloxacin/moxifloxacin)**
  - **linezolid**

    - **clofazimine**
    - **cycloserine**
- **IV agents:**
    - **amikacin**, streptomycin
    - **carbapenem with amoxicillin–clavulanate**
- ⭐ **≥ 5 drugs in the intensive phase; ≥ 4 in continuation.**
- **Surgery** can be beneficial.
- *[added]* **Definitions:**
    - **MDR-TB** = resistant to **isoniazid + rifampicin**
    - **pre-XDR** = MDR + a fluoroquinolone
    - **XDR** = pre-XDR + **bedaquiline or linezolid**
    - WHO now favours all-oral **BPaLM** (bedaquiline, pretomanid, linezolid ± moxifloxacin) for **6 months**

2.12 Extrapulmonary TB Treatment (Slide 31) ⭐

Form	Regimen
Lymph node, pleural, miliary, pericardial, adrenal, GI, GU	**Standard therapy** (6 months)
⭐ **TB meningitis**	**2 months 4-drug intensive + 7–10 months INH + rifampicin** (≈ **9–12 months**) + ⭐ **adjunctive dexamethasone or prednisolone for 6–8 weeks**
⭐ **Pott disease / bone and joint**	**≥ 9 months**; surgery may be needed

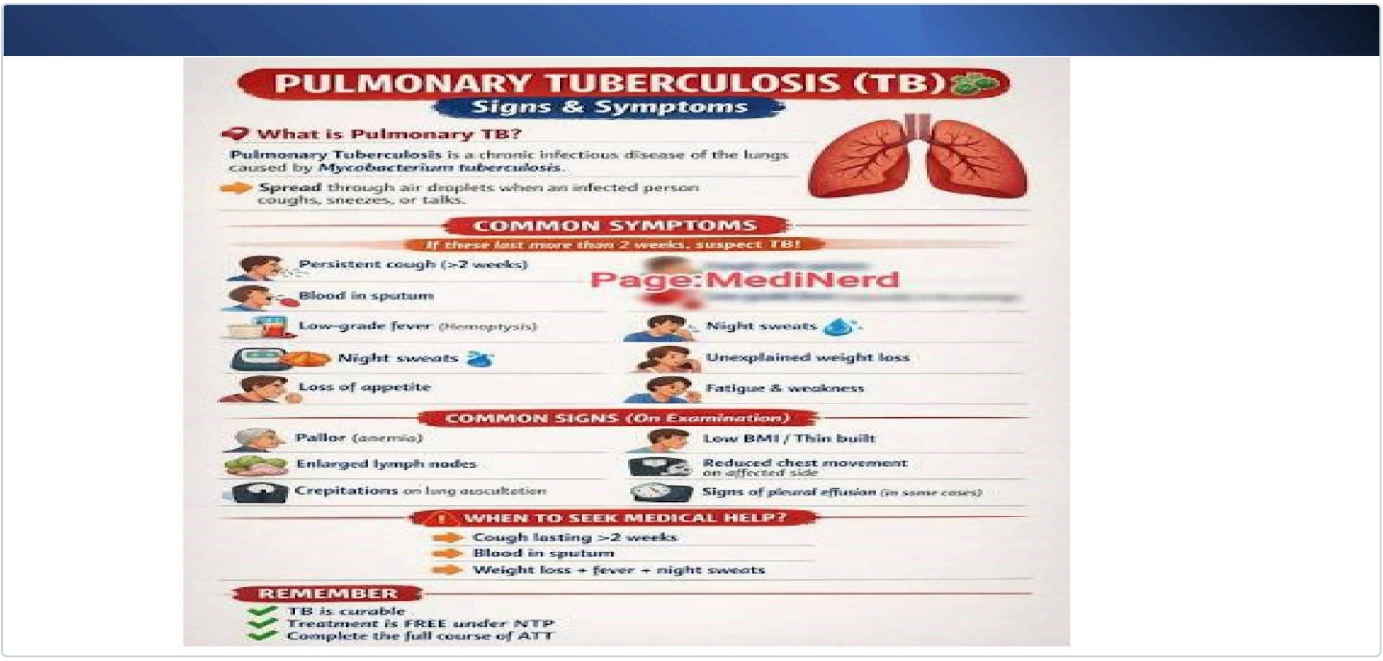
[added] Steroids are also used in **TB pericarditis** in some guidelines (reducing constriction); they are not routine in HIV.

2.13 Latent TB Treatment (Slide 32) ⭐

Indication: positive IGRA or TST after active TB has been excluded. ⭐ Rifamycin-based short regimens are preferred.

Regimen	Details
⭐ **3HP**	**Once-weekly isoniazid + rifapentine for 3 months** (12 doses) + pyridoxine
3HR	**Once-daily isoniazid + rifampicin for 3 months** + pyridoxine
[added] **4R**	Daily **rifampicin for 4 months** (preferred, no INH hepatotoxicity)
[added] **6–9H**	Daily isoniazid 6–9 months (alternative)

⚠️ The slide pairs the **daily** 3HR regimen with "**once-weekly** pyridoxine". With daily isoniazid, pyridoxine is normally given **daily**; treat this as a slide error.



Slide 33: Summary infographic (pulmonary TB signs and symptoms)

🔁 Recap: Treatment.

- Notify, airborne isolation, contact tracing.
- **RIPE 2 months → RI 4 months** + B6. DOT; restart if a gap ≥ 14 days in the intensive phase.
- **4-month HPZM** if ≥ 12 years, ≥ 40 kg, not pregnant, no QT issue.
- **MDR:** ≥ 5 drugs then ≥ 4 (FQ, linezolid, clofazimine, cycloserine, amikacin, carbapenem).
- **TB meningitis:** 9–12 months + steroids. **Pott:** ≥ 9 months.
- **Latent:** 3HP or 3HR (rifamycin-based preferred).

2.14 Approach to Haemoptysis *[added: the lecture title's theme, not covered on the slides]*

Category	Causes
Infection	⭐ **TB**, bronchiectasis, lung abscess, pneumonia, aspergilloma (in an old TB cavity)
Neoplasm	⭐ **Lung cancer** (smoker > 40), carcinoid
Vascular	**PE**, mitral stenosis, AV malformation, LVF
Vasculitis	Granulomatosis with polyangiitis, Goodpasture's, SLE
Other	Anticoagulants, trauma, foreign body

- **Massive haemoptysis** (> 200–600 mL/24 h): **lie on the bleeding side**, secure the airway, CT angiography → **bronchial artery embolisation**.

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3. High-Yield Comparison Tables

Table A: Primary vs Reactivation vs Miliary TB

	Primary	Reactivation	Miliary
Lesion	**Ghon complex** (focus + hilar node)	**Upper-lobe cavity**, caseation	Diffuse 1–3 mm nodules
Zone	Mid/lower	**Upper** (O_2)	Diffuse
Host	First infection	Waning immunity	Immunodeficient (HIV, malnutrition), children
Clue	Hilar node	Haemoptysis, cavity	**Choroidal tubercles**, HSM

Table B: TB Treatment Durations

Situation	Duration
Drug-susceptible pulmonary (RIPE)	**6 months** (2 + 4)
HPZM regimen	**4 months** (8 + 9 weeks)
TB meningitis	**9–12 months** + steroids
Bone/joint (Pott)	**≥ 9 months**
Latent (3HP / 3HR)	**3 months**
[added] Latent (4R)	4 months
Interruption ≥ 14 days in the intensive phase	**Restart**

Table C: TB Specimens

Site	Specimen
Lung	**≥ 3 sputum** (≥ 1 early morning), induced sputum, **BAL**
Child who can't expectorate	**Gastric lavage** (*[added]* early morning)
Pleura	Fluid (**lymphocytic exudate, ADA ↑**) ± biopsy
Lymph node	**Biopsy** (caseating granuloma)
Skin	Biopsy (histology, AFB, culture)
CSF/pericardium/peritoneum	Fluid: cells, chemistry, AFB, **ADA**

Table D: Extrapulmonary TB at a Glance

Form	Hallmark
Lymphadenitis	**Cervical; cold abscess** (commonest extrapulmonary)
Pleural	**Hypersensitivity**; lymphocytic exudate
Miliary	**Choroidal tubercles**, HSM
Pericardial	**Constrictive pericarditis**
GI	**M. bovis; ileocaecal**
GU	**Sterile pyuria**, resistant UTIs, haematuria
Pott	**Kyphosis/gibbus, paraplegia**, cold abscess
Meningitis	Basal; CN VI; low glucose, high protein; **steroids**
[added] Adrenal	**Addison's** (10–20% of primary AI)

4. Rules, Exceptions, Patterns

- **Primary TB → mid/lower zones + hilar nodes; reactivation → upper lobes** (O_2 tension).
- **> 90% of infections are controlled (latent); < 10% progress.**
- **M. bovis = milk = GI TB.**
- **Culture = gold standard; NAAT = rapid** (± rifampicin resistance).
- **3 sputum samples, ≥ 1 early morning.**
- **TST is confounded by BCG → use IGRA**, except in children **< 2 years** (TST).
- **Neither TST nor IGRA distinguishes latent from active** → a positive test requires a **CXR**.
- **RIPE 2 months → RI 4 months; B6 with isoniazid.**
- **Interruption ≥ 14 days in the intensive phase → restart.**
- **4-month regimen excluded in:** < 12 years, < 40 kg, pregnancy, QT prolongation, interactions.
- **MDR: ≥ 5 drugs intensive, ≥ 4 continuation.**
- **TB meningitis (and pericarditis) → steroids**; meningitis and bone TB need **longer** courses.
- **Latent TB: treat only after excluding active TB**; rifamycin-based short courses preferred.
- **TB is notifiable**; airborne precautions; contact tracing.

5. Clinical Correlations

- **Endocrine:**
  - **TB adrenalitis** → Addison's (Adrenal II)
  - **rifampicin induces cortisol metabolism** → double steroid doses in Addison's patients *[added]*
- **Cardiology:** TB **constrictive pericarditis** (Pericardial lecture): calcified pericardium, Kussmaul's sign.
- **Pleura: lymphocytic exudate, ADA > 40 U/L** *[added]*, low glucose; **pleural biopsy** has the best yield.
- **Nephrology: sterile pyuria**; CKD raises TB risk; **ethambutol dose by renal function**.
- **HIV:** the strongest risk factor for reactivation; atypical CXR; **rifampicin–ARV interactions** *[added]*.
- **Rheumatology:** screen for **latent TB before anti-TNF** biologics *[added]*.
- **Haemoptysis work-up in a smoker:** TB vs **lung cancer** vs bronchiectasis.
- **Gastroenterology: ileocaecal TB vs Crohn's**: biopsy, culture, IGRA before steroids or biologics.
- **Public health:** notification, contact tracing, DOT, airborne isolation.

6. Rapid Revision (Last-Day)

1. **M. tuberculosis complex:** M. tuberculosis (droplet, humans), **M. bovis (milk, cattle)**, M. africanum (Africa), M. microti.
2. **Risks:**
   - health-care work, migration/travel, **close contact**, prisons, homelessness, smoking
   - **HIV, CKD, DM**
   - India has the highest burden (slide: 21%, 2009 data)
3. **Primary:**
   - droplet → alveolar macrophage (survives) → bacteraemia

- caseating granuloma → **Ghon focus (mid/lower) + node = Ghon complex** → fibrosis/calcification (latent)

4. **Deficient immunity** → progressive primary → **miliary**. > 90% controlled / < 10% progress.
5. **Reactivation: upper lobes** (O_2), caseating granulomas, **Langhans giant cells**, **cavitation**.
6. **Symptoms:** low-grade fever, **night sweats**, **weight loss**, cough → purulent → **haemoptysis**, dyspnoea, pleuritic pain.
7. **Signs:** pallor, clubbing, wasting; consolidation (dull, crackles); **cavity (hyperresonant, bronchial breathing)**; effusion; **erythema nodosum**, conjunctivitis.
8. **Extrapulmonary:**
   - **nodes** (firm → fixed → **cold abscess**)
   - **pleural** (hypersensitivity)
   - **miliary** (**choroidal tubercles**, HSM)
   - **pericardial** (effusion, **constriction**)
   - **GI** (milk/sputum; ileocaecal)
   - **GU** (haematuria, storage symptoms, HTN, **resistant UTIs**)
   - **Pott** (vertebral body → disc collapse; kyphosis; paraplegia)
9. **Diagnosis:**
   - **AFB smear + NAAT** → **culture (confirmatory)** → DST
   - CXR/CT
   - **3 sputum 8–24 h apart, ≥ 1 early morning**
   - induced sputum, gastric lavage, BAL
   - extrapulmonary: fluid/tissue + **ADA**
10. **Labs:** leukocytosis, anaemia, **hyponatraemia**, ↑ ESR, pyuria, baseline LFTs, **IGRA**.
11. **Algorithm:** - no symptoms → latent → TST (BCG confounds) / IGRA (not < 2 years) → positive → exclude active TB - symptoms → CXR + 3 sputum → microscopy → culture ± molecular → DST
12. **Principles:** cure, ↓ transmission, **prevent resistance**. **Notify; airborne precautions (mask the patient, respirators for staff); contact tracing.**
13. **RIPE: 2 months RIPE + 4 months RI**; **pyridoxine**; daily; DOT; **restart if a gap ≥ 14 days** in the intensive phase.
14. **4-month regimen:** rifapentine + INH + PZA + **moxifloxacin** 8 weeks → rifapentine + INH + moxifloxacin 9 weeks. Requires ≥ 12 years, ≥ 40 kg; not pregnant; no QT prolongation.
15. **MDR:** FQ, linezolid, clofazimine, cycloserine; amikacin, streptomycin, carbapenem + co-amoxiclav; **≥ 5 then ≥ 4 drugs**; ± surgery.
16. **Meningitis:** 2 months + **7–10 months** + **dexamethasone/prednisolone 6–8 weeks**. **Bone/Pott: ≥ 9 months** ± surgery.
17. **Latent:** after excluding active TB. **3HP weekly × 3 months** or **3HR daily × 3 months** + B6.

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7. Exam-Style Questions

Q1 (lecture case 1, slide 34). A 48-year-old mechanic has a **pre-placement TB test** before **volunteering at a refugee clinic**. He has had a mild cough, sore throat and headaches for 1 week. Background: HTN, GERD; **never travelled**, **no TB contact**; smoker (27 pack-years). Vitals are normal and the lungs are clear; WBC 6200; COVID negative. **TST 13 mm at 50 h.** Next step? **A. Chest X-ray** · B. No further management · C. IGRA · D. Repeat TST in 8–10 weeks · E. Isoniazid + rifapentine

> **Answer: A** *[reasoning added; key not on slides]*.
>
> - He is entering a **high-risk congregate health-care setting**, so the CDC cut-off is **≥ 10 mm** and **13 mm is positive**.
> - A positive TST can't distinguish latent from active TB, so **exclude active TB with a CXR** before any latent-TB treatment (slide 26: "if TST or IGRA positive, exclude active TB").
> - Why not the others:
> - **E** is premature until the CXR is clear.
> - **D** (two-step TST) is for an initially **negative** baseline test.
> - **B** is wrong because 13 mm is positive here.
> - ⚠️ Nuance: CDC 2019 health-care-worker guidance allows a **confirmatory second test (IGRA)** for a positive result in a **low-risk asymptomatic** worker. If an answer key says **C**, that is the logic. The standard board answer for a positive TST is **CXR**.

Q2 (lecture case 2, slide 35). A 7-year-old **BCG-vaccinated** boy arrived from **Ukraine (high TB burden)** 2 months ago. He is asymptomatic (only allergic rhinitis-type symptoms) with a normal examination. Next step? A. TST · B. No further evaluation · C. CXR · D. Sputum microscopy and culture · **E. IGRA**

> **Answer: E** *[reasoning added; key not on slides]*.
>
> - A recent immigrant from a high-burden country needs **screening for latent TB**.
> - He is **BCG-vaccinated and ≥ 2 years old** → **IGRA** avoids BCG false positives (slide 26: TST "more difficult to interpret if previous BCG"; IGRA "not if < 2 years").
> - CXR/sputum are for **symptomatic** patients or after a positive screen.

Q3. A 35-year-old with HIV has 6 weeks of fever, night sweats, 8 kg weight loss and cough with blood-streaked sputum. CXR shows a right upper-lobe cavity. Best initial microbiological approach? A. Single sputum AFB smear · **B. Three sputum samples (≥ 1 early morning) for AFB smear and NAAT, with culture and susceptibility testing** · C. IGRA · D. TST · E. Bronchoscopy first

> **Answer: B** (slides 22, 25, 26).
>
> - Active TB needs ≥ 3 samples with **AFB + NAAT**, then **culture** (the gold standard) and **DST**.
> - IGRA/TST can't diagnose active disease.
> - Bronchoscopy/BAL is used if sputum is non-diagnostic.

Q4. A patient in the intensive phase of RIPE stops all medication for 3 weeks because of nausea, then returns. Management? A. Continue where he left off · B. Extend the continuation phase by 3 weeks · **C. Restart the full treatment course** · D. Switch to the 4-month regimen

> **Answer: C** (slide 28): an interruption of **≥ 14 days during the intensive phase → restart full treatment**.

Q5. A patient on RIPE develops tingling feet. Two months later he reports blurred vision and difficulty distinguishing red from green. Most likely culprits? **A. Isoniazid (neuropathy) and ethambutol (optic neuritis)** · B. Rifampicin and pyrazinamide · C. Pyrazinamide and ethambutol · D. Rifampicin and isoniazid

Answer: A *[added pharmacology]*.

- Isoniazid → B6-related neuropathy (prevented with **pyridoxine**, slide 28).
- Ethambutol → optic neuritis, red–green colour blindness.

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8. Exam Pitfalls & Traps

Trap	Correct thinking
Ghon focus in the upper lobe	❌ **Primary Ghon focus = mid/lower zones**; **reactivation = upper lobes**
Positive TST/IGRA = active TB	❌ Neither distinguishes latent from active → **CXR** (± sputum)
Treat latent TB immediately after a positive test	❌ **Exclude active TB first** (otherwise monotherapy breeds resistance)
TST in a BCG-vaccinated adult	⚠️ False positives → prefer **IGRA** (≥ 2 years)
IGRA in a 1-year-old	⚠️ Slide: IGRA "not if < 2 years" → TST
One sputum sample is enough	❌ **≥ 3**, 8–24 h apart, ≥ 1 early morning
Smear negative = no TB	❌ Smear is insensitive; **culture** (gold standard) / NAAT
Continue RIPE after a 3-week gap in the intensive phase	❌ ≥ 14 days → **restart**
Forget pyridoxine	❌ **B6 with isoniazid** (neuropathy)
4-month HPZM in a pregnant or QT-prolonged patient	❌ Contraindicated (moxifloxacin)
6 months for TB meningitis	❌ **9–12 months + steroids**
6 months for Pott disease	❌ **≥ 9 months** ± surgery
MDR-TB with 3 drugs	❌ **≥ 5 intensive, ≥ 4 continuation**
TB isolation = surgical mask for staff	❌ **Respirators (N95) for staff**; surgical mask for the **patient**
Not reporting TB	❌ **Mandatory notification** + contact tracing
GI TB only from swallowed sputum	⚠️ Also **M. bovis in unpasteurised milk**
Recurrent "UTI" with sterile cultures	⚠️ Think **genitourinary TB**
Slide 18 = TB meningitis features	⚠️ Slide error: the content is **pericardial TB**
Global burden 9.4 M / India 21%	⚠️ 2009 data; now ~10.8 M, India ~26% (WHO 2024)